

CHRISTOS PATRINOS

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KEY ACHIEVEMENTS

- Developed simulation methodologies across the automotive and aerospace sectors at Aston Martin Lagonda and Raytheon.
- Developed **SpearDAT** as an independent engineering project: a modern C++17/MATLAB missile aerodynamic analysis framework based on publicly released Missile DATCOM methods, supported by comprehensive software verification and validation against NASA wind-tunnel datasets.
- Developed **VDSE** (Vehicle Dynamics Simulation Environment) using MATLAB/Simulink and Python, benchmarking its performance against available open-source vehicle dynamics solvers.

CAREER HISTORY

Raytheon UK, Harlow, UK

Sep. 2024 – Present

Senior Systems Engineer, Modelling & Simulation

TEAM: Future Systems

- Developing high-fidelity 6DOF flight dynamics and component-specific simulation models in MATLAB and Simulink to support system design and analysis.
- Implementing guidance and control algorithms within high-fidelity simulation models.
- Conducting large-scale performance assessment studies to define and optimise operational system envelopes in line with mission requirements.
- Engaging directly with customer and end-user stakeholders at operational sites to understand requirements, present technical work and ensure simulation activities address operational needs.
- Contributing to model-based development and software-in-the-loop verification of Mission Control Unit software against system requirements.
- Developing MATLAB applications and tools for mission planning and preliminary performance analysis.
- Leading model correlation activities using flight-trial data to validate and refine simulation fidelity against real-world performance.
- Working within Agile/Scrum delivery, using Jira for sprint planning and work tracking and Bitbucket/Git for collaborative version-controlled development.

Aston Martin Lagonda Ltd, Gaydon, UK

Jun. 2021 – Sep. 2024

Chassis Simulation Engineer

TEAM: Vehicle Dynamics and Controls CAE (R&D)

- Led research into simulation methodologies, improving team simulation capabilities using MATLAB, Simulink, Python and Simscape.
- Supported model correlation activities across the full cycle, from experimental validation planning to simulation-led investigation of vehicle dynamics issues.
- Designed and developed MATLAB tools and applications for data management and analysis, adopted by multiple teams across the company.
- Conducted vehicle dynamics performance analysis using objective test data and subjective driver assessments from Aston Martin and competitor vehicles, identifying performance trends to inform future vehicle development.
- Implemented control system algorithms supporting vehicle dynamics enhancements in real-time simulation environments.
- Coordinated driver-in-the-loop sessions with professional drivers for lap-time performance assessment.
- Collaborated with the powertrain team to develop integrated chassis and powertrain simulation models.

Cranfield University, Cranfield, UK

Oct. 2020 – May 2021

Research Assistant

GROUP: Advanced Vehicle Engineering Centre

- Developed VDSE, a research vehicle dynamics simulation tool with 3D animation, combining MATLAB/Simulink and Python.
- Formulated the mathematical model of a novel hydropneumatic suspension for real-time vehicle dynamics simulation.
- Developed an optimisation strategy for suspension parameters using large-scale design experiments.
- Analysed the hydropneumatic suspension's low-level control system.
- Supported teaching activities in Vehicle Dynamics and Control.

Thenamaris (Ships Management) Inc., Athens, Greece

Apr. 2019 – Sep. 2019

Project Specialist

DEPARTMENT: Energy Performance

- Streamlined propeller performance evaluations using Python.
- Developed a Python-based method for estimating vessel NOx emissions.

Rapidproto Hellas LLC, Chalkida, Greece

Sep. 2018 – Feb. 2019

Model-Based Software Developer

- Developed application software using MATLAB, Simulink, Stateflow, TargetLink and SCADE for air-suspension systems of a prototype SUV.

EDUCATION

Massachusetts Institute of Technology (MIT)

Aug. 2023 – Aug. 2024

MicroMasters in Statistics and Data Science

- Completed courses in Probability Theory (91%), Fundamentals of Statistics (86%), Data Analysis and Statistical Modelling (96%) and Machine Learning with Python (95%).

Cranfield University, Cranfield, UK

Oct. 2019 – Sep. 2020

MSc Automotive Engineering

- Distinction, 81%; ranked within the top 2 of a 44-student cohort.
- Specialised in Vehicle Dynamics and Control Systems.
- MSc Thesis: *Integrated Chassis Control of a High-Performance Vehicle* - 83% (Excellent).

National Technical University of Athens, Athens, Greece

Sep. 2012 – Sep. 2017

MEng Mechanical Engineering

- Final grade: 7.9/10.0 (Very Good); top 10% of a 180-student cohort.
- Specialised in Mechanical Design.
- Thesis: *Investigation and improvement of the operation of a prototype SLS/SLM machine* - 10.0/10.0 (Excellent).

SKILLS

Programming

MATLAB, Python, C++

Modelling & Simulation

Simulink, Simscape, VI-CarRealTime, MSC Adams Car

Development Workflow

Git, Bitbucket, Sourcetree, Azure DevOps, Jira; Agile/Scrum

Languages

Greek (native), English (fluent)